1 Physics Period 1, Week 5

1. You toss an apple horizontally at 8.7 m/s from a height of 2.6 m. Simultaneously, you drop a peach from the same height. How long does each take to reach the ground?

2. A carpenter tosses a shingle horizontally off an 8.8-m-high roof at 11 m/s.

   (a) How long does it take the shingle to reach the ground?

   (b) How far does it move horizontally?

3. **Bonus**

   As an expert witness, you're testifying in a case involving a motorcycle accident. A motorcyclist driving in a 60-km/h zone hit a stopped car on a level road. The motorcyclist was thrown from his bike and landed 39 m down the road. You're asked whether he was speeding. What's your answer?

4. **Bonus**

   Standing on the ground 3.0 m from a building, you want to throw a package from your 1.5-m shoulder level to someone in a window 4.2 m above the ground. At what speed and angle should you throw the package so it just barely clears the windowsill?